

TEST009 – Recommendation Engine Functional Testing

Steps and Tests Performed:

1. Created a new test branch from main.
2. Developed a test script (scripts/test_recommendation_engine.js) to run the backend recommendation engine with valid sample inputs.
3. Ran the script using Node.js to simulate recommendations for a sample product and user profile.
4. Verified that the recommendations matched the acceptance criteria for filtering and logical ranking.

Test Inputs:

- Original Product: Milk Chocolate (contains milk, high sugar/fat)
- User Profile: Vegan, allergic to milk, avoids additive 621
- Candidate Products: Dark Chocolate, Fruit Bar, White Chocolate

Results:

- Recommendations Returned:

1. Dark Chocolate (Score: 95, Safety: green, Reasons: Healthier, Safe, Vegan)
2. Fruit Bar (Score: 67.5, Safety: green, Reasons: Healthier, Safe)

- Filtering: Products with allergens or not matching dietary preferences were filtered out as expected.

- Scoring: Results were logically ranked by healthiness and user safety, with the highest scoring and safest products at the top.

Changes Made:

- Added a new test script for functional backend testing.
- No changes made to the main code.

Errors Found:

- No errors or issues were encountered during the test run. The engine performed as expected.

This document is provided for sign-off as per instructions.